BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Algorithm Design and Flowcharts

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

1. What is abstraction?

2. What does WHERE do in SQL?
   A) Names the source table
   B) Filters rows
   C) Chooses output columns
   D) Sorts all rows

3. A zero-indexed 2D array is [[11,12],[21,22]]. Give grid[1,0], then state the indexes needed to access 12.

4. Two mono recordings have the same sample rate and bit depth. One lasts twice as long. Does this double its quality? How does raw size change?

Today's objectives

Recall the essential ideas of algorithm design and flowcharts. Apply a clearly modelled method.

Key vocabulary - Read, cover, recall

Flowchart
Diagram representing algorithm steps and control flow.

Decision
Diamond that selects between paths using a condition.

Process
Rectangle representing an operation.

Input/output
Parallelogram for reading or displaying data.

BLACK PEN: Write one definition in your own words.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New Learning - Key ideas

Clear algorithms
An algorithm must be finite, precise and able to produce an intended result for allowed inputs.

Basic flowchart shapes
Use terminal (start/end), process (rectangle), input/output (parallelogram), decision (diamond) and labelled arrows.

Selection and iteration
A decision directs execution along yes/no paths; an invalid input can loop back to the input step.

Equivalent pseudocode
Trace both representations on the same input to check they perform the same actions.

I Do - Worked model

Design a flowchart that repeatedly asks for a mark until it is between 0 and 100 inclusive. [5 marks]

- Establish the input
- START -> input mark using a parallelogram.
- Decision checks whether 0 <= mark <= 100; YES continues to END.
- NO returns to input mark, rather than ending; label both arrows. [5]

We Do - Guided practice

1. Write the steps for an algorithm that asks for an age until age >= 0. [4]

2. Explain the difference between a process symbol and decision symbol. [4]

3. Trace a validation flowchart when marks entered are -1, 75. [4]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Correct the Boolean operator used for checking an IPv4 octet.

```
octet = 260
if octet >= 0 OR octet <= 255 then
print("Valid range")
else
print("Invalid")
endif
```

You Do - Independent exam questions

Q2. State the shape normally used for a decision in a flowchart. [1]

Q3. An algorithm must accept an integer from 1 to 10. Describe the paths for 0 followed by 6. [4]

Q4. Explain one reason for using a trace table while refining an algorithm. [3]

Exit Ticket - True or False?

A trace table can help check how values change during a program.

TRUE [ ] **FALSE [ ]**

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick each correct point; correct mistakes.

Do Now - Model answers

1. Removing irrelevant detail while keeping what is needed to solve the problem.

2. B - Filters rows

3. grid[1,0] is 21; 12 is at grid[0,1].

4. Raw size doubles: there are twice as many samples. Quality does not improve. Sample rate and bit depth stay the same.

I Do - Explain the model

Establish the input

START -> input mark using a parallelogram.

Decision checks whether 0 <= mark <= 100; YES continues to END.

NO returns to input mark, rather than ending; label both arrows. [5]

We Do - Model answers

1. Start/input age [1]; decision age >= 0 [1]; valid branch ends/continues [1]; invalid branch loops to input [1].

2. Process is a rectangle [1] representing an operation/calculation [1]; decision is a diamond [1] selecting a path depending on a condition [1].

3. -1 is rejected as below range [1]; arrow loops back [1]; 75 is within 0-100 [1]; flow follows valid branch to end [1].

Code First - Answer

```
octet = 260
if octet >= 0 AND octet <= 255 then
print("Valid range")
else
print("Invalid")
endif
```

Expected output: Invalid

You Do - Self-mark

Q2: Diamond [1].

Q3: 0 fails validity [1] and loops to input [1]; 6 passes [1] and algorithm continues/exits [1].

Q4: Records changes in variables [1] for each relevant step [1] to identify unexpected logic/errors [1].

GREEN PEN: Choose the hardest answer. Correct it on lined paper.